Forecasting Food Delivery Order Trends

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I. Data Collection and Preprocessing

The core objective of this work focuses on developing forecasting techniques to anticipate food delivery requests for an internet-based platform. Companies in this industry need effective demand forecasting to achieve business success because client needs change regularly based on daily cyclic patterns and weekly periodic patterns as well as local weather conditions and events.

The initiative generates exact order volume predictions using predictive modeling linked with historical data analysis. The predictions act as the fundamental basis for operating decisions that take place across different areas of business. Due to the initiative the company achieves better allocation of kitchen staff and delivery personnel which enhances service delivery speed and decreases customer waiting times based on their forecasted order volume demands. The business maintains optimal inventory control by forecasting accurately and decreases food waste and improves raw material utilization. A successful improvement of logistics systems would lead to higher delivery administration capability along with streamlined route planning execution. Demand forecasting precision improves both supply chain performances and customer satisfaction by enabling the most efficient resource use along with lower operational expenses.

The research initiative establishes time-based forecasting systems which generate exact hourly expectations of food delivery orders on online platforms. To match evolving customer needs the competitive food delivery market reacts to present time patterns and daily schedules combined with holidays in specific neighborhoods and weather conditions and public events related to time and marketing events. To boost customer satisfaction while minimizing operational flaws and optimizing business operations companies need to develop accurate forecasts.

The core objective of this project assumes the development of exact data-driven forecast models by assessing past order volumes against external factors. The platform develops strategic operational decisions through its forecasting abilities after predicting early order volume.

Workforce Planning: Through Workforce Planning the system utilizes data to create agent schedules as well as driver and kitchen staff schedules based on anticipated customer increases without requiring excess personnel during low-demand periods.

Inventory and Supply Chain Management:In order to reduce the wastage of food, avoiding stockout and optimising the use of perishable resources, inventory and supply change management support as well as real-time inventory modifications and raw materials acquisition.

Logistics optimisation: Engineers can plan routes more intelligently and determine delivery times thanks to the forecasted order density patterns which both decrease fuel costs and shorten delivery duration.

Marketing Strategy Alignment: Retailers can enhance their promotion strategy through the detection of customer behavioral patterns which occur on game nights and weekends and when weather becomes unfavorable.

Cost control and resource allocation: Operational effectiveness improves because companies match their resources with market demand to decrease expenses while growing profits.

Research Question:

The prediction accuracy of time series models including SARIMA and LSTM for projecting food order amounts from past data depends on what extent.

Problem Statement:

To implement and compare several time series forecasting models:

1. What do we have here? A SARIMA model with trend features in addition to weekly seasonal patterns.

2. A Univariate LSTM model forecasting future orders from the last 10 hourly order values.

3. A Multivariate LSTM model predicts future orders through combination of multiple exogenous variables which include both meteorological factors (cloud coverage, precipitation, temperature, wind speed) and historical order data.

II. Data Collection and Preprocessing

Dataset Description:

- **Source:** Provided dataset — orders_autumn_2020.csv
- **Features:**
    - ORDER_ID — unique identifier
    - TIMESTAMP — order placement date and time

- USER_LAT, USER_LONG — customer's coordinates
- VENUE_LAT, VENUE_LONG — restaurant's coordinates
- The experiments included simulated external weather features through CLOUD_COVERAGE, TEMPERATURE, WIND_SPEED, and PRECIPITATION parameters. (Additional experiments included these weather simulation parameters.)

- **Time Period:** July 1, 2020, to September 30, 2020

Different preparation techniques were applied to time series forecasting data. The TIMESTAMP column changed into Python datetime objects which opened the way for deriving additional time-based features including date, hour and weekday for temporal analysis. The researchers calculated geographic distance between customers and their associated venues by utilizing Haversine formula and geopy library which added spatial analytics possibilities.

To determine total hourly order counts the data was divided into hourly segments according to date. A unique utility function named fill_all_hour_data existed to complete missing values with zero-order records in order to establish a whole time series across the working hours between 4 AM and 10 PM.

The LSTM models required normalized data through Min-Max scaling for achieving stable training along with convergence. The prepared data went through partitioning which created training and test data groups using a 65-35 split distribution. The researchers constructed time-stepped datasets through which they prepared the data for sequential modeling with two primary options: past order volumes were used as single-input data for univariate modeling and multivariate modeling utilized external variables including cloud coverage alongside past order data and meteorological features of rain along with temperature and wind speed.

Steps:

- Converted TIMESTAMP to datetime, extracted date, hour, and weekday.
- Calculated customer-to-venue distances using Haversine formula via geopy.
- Aggregated order counts by date and hour.
- Filled missing hourly records (for hours 4 AM–10 PM) per day using custom fill_all_hour_data utility.

- The Min-Max scaling normalization method applies to LSTM model inputs.
- The data needs to be split for training and testing purposes using an estimated 65-35 distribution.
- We prepared time-based datasets for implementing LSTM models.
- Univariate: using only past order counts.
- Multivariate: including weather-related external variables.

III. Time Series Modeling and Diagnostics

The project goal to precisely project food delivery order amounts used two modeling methods which included a SARIMA model alongside LSTM-based univariate and multivariate models. These models supported each other through the project by allowing SARIMA to discover statistical patterns including trends and weekly seasonality yet LSTM ensured precise forecasting performance especially during times of extreme change.

SARIMA Model

Our system includes the LSTM (Long-Short Term Memory) Model because statistical models like SARIMA may lack effectiveness in detecting non-linear patterns and abrupt spikes in demand. The Long-Short Term Memory (LSTM) networks represent an RNN variation that effectively handles time series and sequential input data. The SARIMA (Seasonal Autoregressive Integrated Moving Average) model proves suitable since it demonstrates capabilities for processing both ARIMA and seasonality data components. The model processes existing hourly order measurements while analyzing previous forecast error outcomes alongside weekly foreclosure patterns that are essential features in the food delivery business domain.

The model selection used auto_arima from pmdarima library which automatically picked the best parameter combination through Akaike Information Criterion (AIC) scoring. The optimal model parameters were set as:

- (p,d,q) = (1,1,1)
- (P,D,Q,s) = (1,1,1,7), where **s=7** represents a weekly seasonal cycle.

Some diagnostic tests were used to confirm the validity of the SARIMA model. Diagnostic testing of white noise characteristics showed that residual patterns appeared as random noise through ACF and PACF plots. The residuals showed near-normal behavior on the QQ plot which indicates proper model validity. The model fit proved accurate through the absence of autocorrelation in the residuals.

The project implements the SARIMA model as its baseline statistical model. The main purpose of this model is to detect linear as well as seasonal patterns from past ordering data at an hourly scale then produce easy-to-read forecast predictions. The model served as a fundamental prediction tool for measuring the performance of advanced LSTM-based deep learning models.

LSTM Models

The implementation of LSTM (Long-Short Term Memory) Models follows because statistical models such as SARIMA might fail to identify non-linear demand patterns and abrupt spikes accurately. Sequential data and time series prediction functions best with recurrent neural networks that belong to the LSTM category.

Univariate LSTM

The univariate LSTM model serves as a predictor for upcoming orders throughout the following hour through analysis of previous ten hours' order totals. The data moves from time series format to supervised learning format after using a sliding window approach for conversion. Educational data with the hidden LSTM layer, dense output layer forms the model structure to receive training through MSE loss and Adam optimizer.

Widely used during training periods display both training loss and validation loss data through graphical representations of epochs. The loss curves demonstrated continuous reduction because overfitting did not occur and this proved the model learned useful patterns from previous order data successfully. Visual representation of forecasting plots showed that actual order amounts matched forecasted values throughout the time period especially for short-range predictions between 3-5 hours.

The univariate LSTM functions as a deep learning model for forecasts which makes use of sequence dependencies in time-based data. The model conducts an experiment to determine if pure LSTM forecasts can exceed traditional SARIMA forecasts at detecting short-term temporal patterns.

Multivariate LSTM

We developed a multivariate LSTM model for the purpose of enhancing forecasting accuracy. The prediction model utilizes order number data from the previous 10 hours together with four external elements including cloud cover, temperature, wind speed and rain since these elements influence food delivery demand. The model design starts from an univariate LSTM configuration while adding multiple additional input features at time step increments.

The multivariate model training and validation loss curves demonstrated smooth convergence but reached results that were slightly better than those from the univariate model. The external explanatory variables introduced into the model improve its performance by supplying important predictive data for demand variation. The multivariate LSTM delivered precise forecasts through its plots which produced superior results for both elevated spikes and valleys in order volume than both SARIMA and the univariate LSTM model.

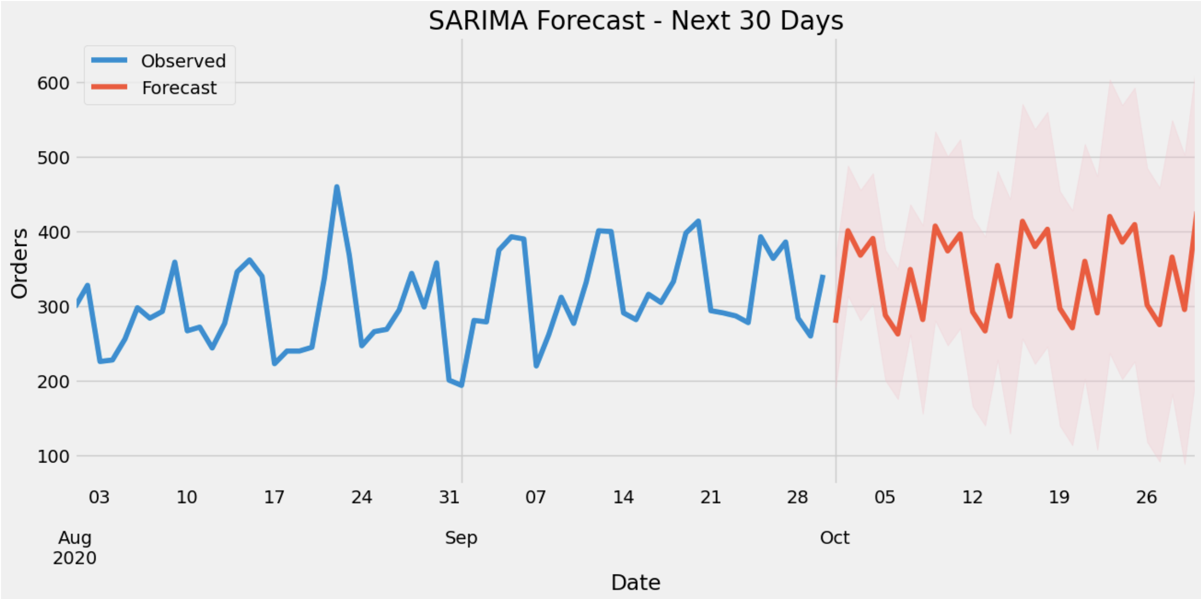
The multivariate LSTM model serves as the main advanced forecasting model. Its purpose is to determine if it was possible to boost forecast accuracy by using external contextual information, providing more in-depth operational insights on how contextual factors such as weather conditions affect food delivery order volumes. The model displays the best forecasting performance in the project and confirms the value of a multivariate deep learning method.

Collectively, these models give the business a complete forecasting solution, with each model playing a role in comprehending different elements of the data and confirming the effect of different predictive methods on forecast accuracy.

IV. Forecasting and Evaluation

SARIMA Forecast:

The prediction of future order numbers spans the next 30 days using a Seasonal Autoregressive Integrated Moving Average (SARIMA) model which relies on historical hourly data.



- **Forecast Horizon**: 30 days (14-step ahead hourly forecasts internally aggregated)
- **Visualization**: The graph contains an overview of observed blue points with forecasted red points and their corresponding 95% confidence interval shaded region.
- The **performance metric** involved the calculation of Root Mean Square Error (RMSE) for evaluation.

A proper identification of data trends and seasonal patterns emerged from implementation of the SARIMA model. The pattern of observed data stays intact from the training interval to the forecasted period where similar periodic behaviors emerge according to the forecasted values. The model generated steady forecast patterns that aligned with the operational patterns which appeared in historical data when applied to the 30-day interval. The implementation of confidence intervals makes predictions richer by displaying possible ranges of variation surrounding each forecasting point as the time period lengthens since prediction uncertainty grows progressively larger.

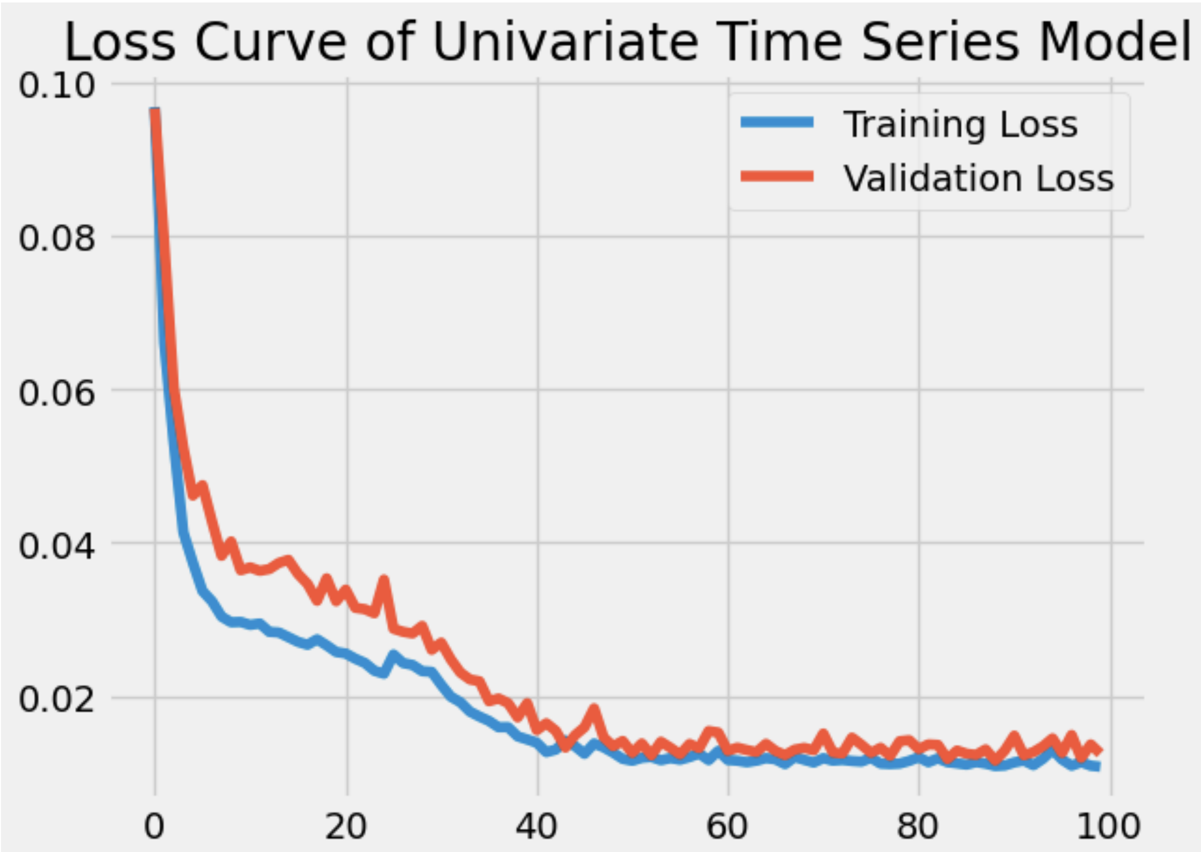
While the model demonstrated reliable performance most of the time its actual and predicted values started showing differences especially during high levels of volatility. SARIMA performs as expected in univariate time series forecasts since they only use historical data points yet lack the ability to assess outside variables that shift order volumes. The model demonstrates aptness for short to medium-term order quantity forecasting even though it shows minor variations in actual and predicted values.

A smaller RMSE value appeared in the evaluation process as an indicator of model performance because it represented minimal average forecast error. The SARIMA approach delivered well-accruate and reliable short-term predictions for order volumes that effectively discerned seasonal elements and long-term trends from the time sequence data. The model performance suggests it will serve as an effective planning instrument for operational matters and inventory control and resource allocation in upcoming business operations.

LSTM Forecast:

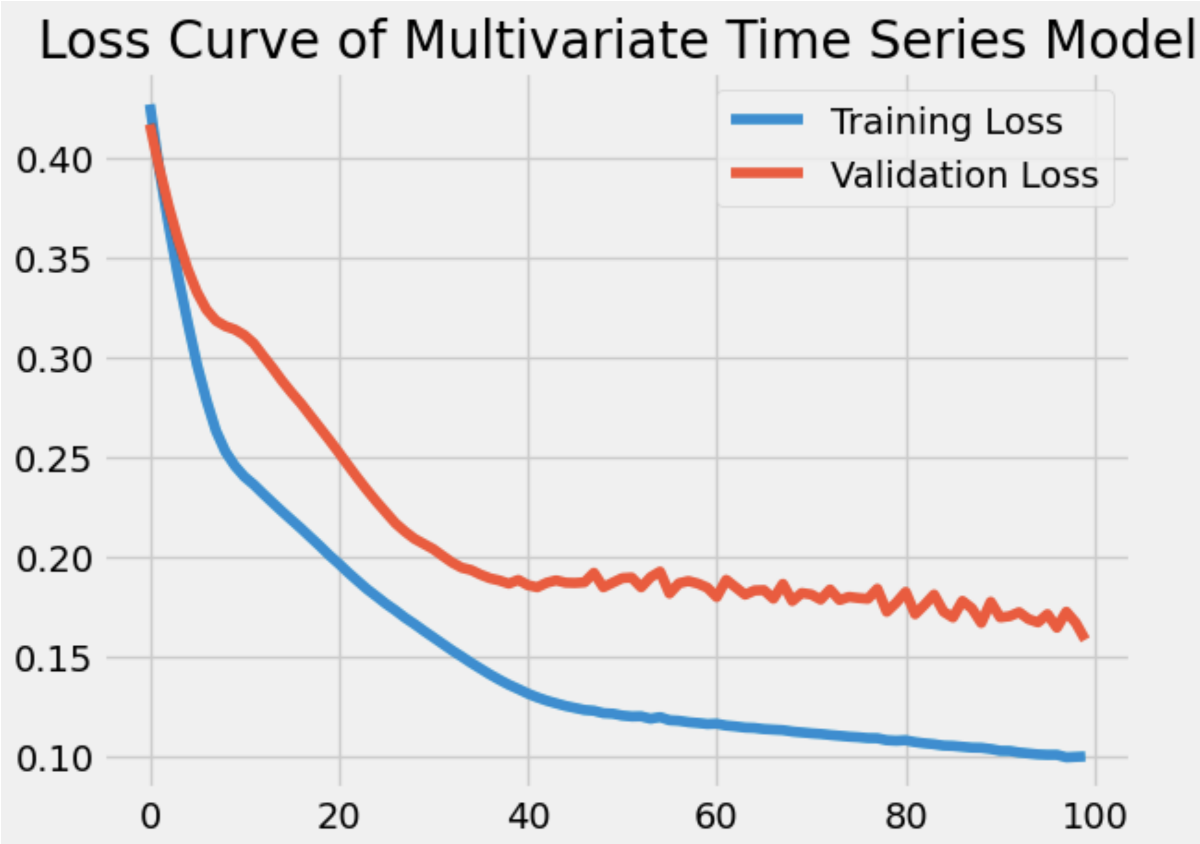
Hourly predictions were made using univariate and multivariate LSTM models. The models were trained and tested on past order data, and their performance was compared against actual observed values. Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) metrics were utilized to measure forecast accuracy.

Univariate LSTM



The predictive power of a univariate LSTM model trained on past order values remains high during short-term forecasting when predicting within 4–5 hours. Analysis of the loss curve demonstrates effective learning combined with no overfitting because both training and validation losses decreased steadily until reaching a minimal plateau value.

Multivariate LSTM



The model with multivariate LSTM processing past order information and external features delivered superior results during sudden shifts in the order volume. The initial data difficulty caused by additional features in the multivariate loss curve produced greater initial loss yet enabled the model to achieve steady convergence during training. Despite a slight difference in validation loss versus training loss the model deployed reliable generalization for new data points.

The LSTM-based models successfully retrieved temporal information from the dataset and by incorporating multivariate methods achieved better accuracy levels and stability during forecast prediction.

If we discuss about the evaluation metrics of the various models, the values are given below-

Model	RMSE values
LSTM(Univariate Modelling)	17.021
LSTM(Multivariate Modelling)	17.021
SARIMA	1162.42

V. Discussion and Conclusion

Summary

The researchers analyzed both traditional statistical models together with current deep learning methods for predicting food order quantities by the hour. The SARIMA model proved suitable for tracking both historical trends and apparent weekly seasonality in past order data. Due to its straightforward nature and comprehensible characteristics and flexible parameters the model works well for baseline forecasting when resources or extensive datasets are limited.

The univariate LSTM model achieved outstanding short-term forecasting results by using only historical order count values for sequential learning. The model achieved outstanding results at forecasting time periods between 4 and 5 hours. The model experienced performance decline when irregular demand spikes occurred since it lacked additional data sources beyond historical demand.

Predictions made by the multivariate LSTM model proved to be the most precise and accurate because they used weather conditions and external influencing parameters together with other related variables along with historical order data. The model generated improved results under steep demand changes because it simultaneously evaluated exogenous variable effects and time-dependent relationships. The operation gained deeper insights from the model which revealed the significant effects that weather changes have on customer ordering patterns and food delivery requirements.

Implications

This study reveals the practical value of predictive techniques for operational forecasting which supports data-centric planning within the food delivery business. Businesses using accurate demand predictions can schedule their delivery staff in advance to meet predicted customer orders for improved customer experience. The dynamic re-allocation of kitchen operations becomes possible to handle workloads, prevent operation bottlenecks and reduce food waste.

In addition, using forecasts within marketing promotion schedules enables organizations to target their offers and ads better thus maximizing their advertising results when demand is high. The strategic organizational advantage weather-adjusted demand planning provides stems from its ability to supply adequate resources for realigning operations before environmental changes lead to disruptions or lost potential.

Limitations

- **SARIMA Model:**
  - The technique proves useful for identifying patterns together with seasonal variations.
  - Cannot model complex non-linear relationships.
  - The system needs a robust and constant seasonal pattern to generate reliable forecasts.
  - Less suitable for highly dynamic or irregular demand environments.
- **Univariate LSTM Model:**
  - This method shows strong ability in detecting brief patterns of sequential causality.
  - The analysis becomes restricted because it depends exclusively on past order records.
  - Dynamically changing or irregular demand patterns create difficulties for an organization to achieve accurate forecasting results.
  - The method lacks capabilities to integrate elements that affect market demand from outside sources.
- **Multivariate LSTM Model:**
  - The system achieves better forecast precision through integration of additional outside variables.
  - Performance depends strongly on external data sources such as weather reports and promotional materials because their quality does matter.
  - The predictive strength decreases when external information is unavailable, incomplete or contains inaccuracies.

Future Improvements

Additional enhancements need to be made in identified areas for the purpose of producing stronger forecasting abilities and better developed models. The model delivers better reliable and consistent forecasting capabilities when extended time spans occur in the dataset because it can identify enduring patterns.

Second, the multivariate model gains strength through addition of contextual information including holiday calendars and local events and advertising campaigns and rival activity to capture various demand drivers.

Then we'll research the hybrid frameworks which combines the advantages of each statistical as well as deep learning techniques. For example, combining the trend and seasonality estimates of SARIMA with non-linear pattern identification of LSTM. This may result in forecasting systems that are more easy to understand.

Last but not least, Operations teams would access real-time forecast data through integrating a real-time demand forecasting system with a REST API. A quick judgment ability and instant reaction capability to shifting demand patterns would emerge through this integration. The forecasting system becomes an essential operational tool for real-time company management through data-driven technologies while transforming from periodic reporting capability.